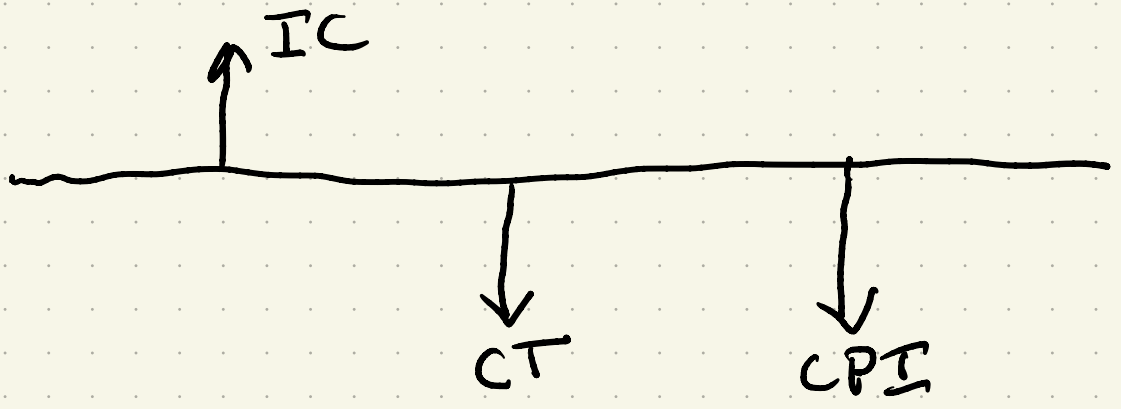
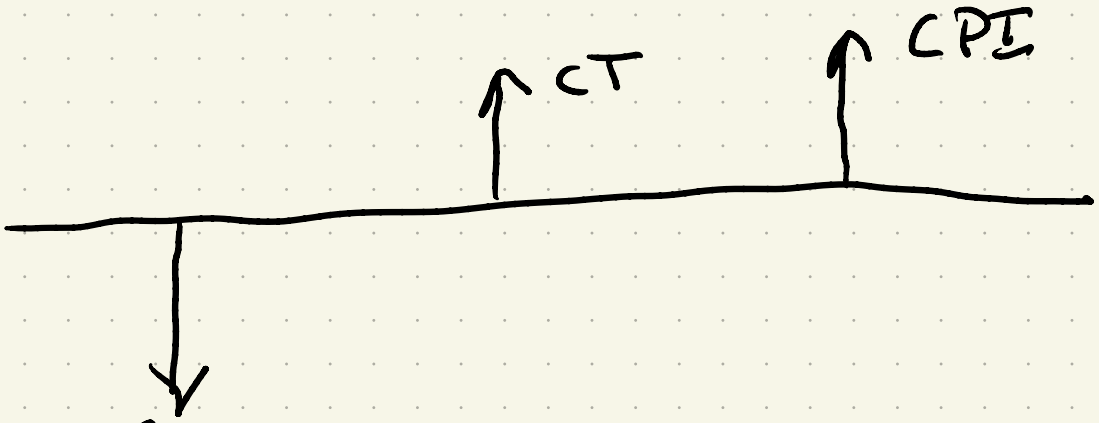


RISC

John Cocke
@IBM 801



CISC

PL/D DEC Vax 11
COPY

MOPS, TOPS

ARM, MIPS, RISCV 100s-1000s of instructions

Speedup

$$\frac{\text{CPU time of } X}{\text{CPU time of } Y}$$

$$\text{Performance} = \frac{1}{\text{CPU time}}$$

X is n times faster than Y

$$n = \frac{\text{Performance of } x}{\text{Performance of } y} = \frac{\text{CPU time } Y}{\text{CPU time } X}$$

Y takes 100s X takes 25s

$$\text{Speedup} = n = \frac{100}{25} = 4 \text{ times}$$

CPU time = IC × CPI × CT

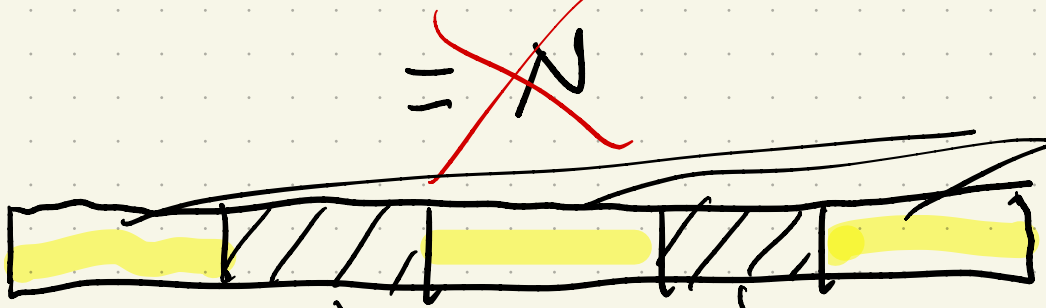
Gene Amdahl IBM 360

1967

Amdahl's law

$$\text{Speedup} = \frac{\text{Execution time on one processor}}{\text{Execution time on N processors}}$$

$= N$

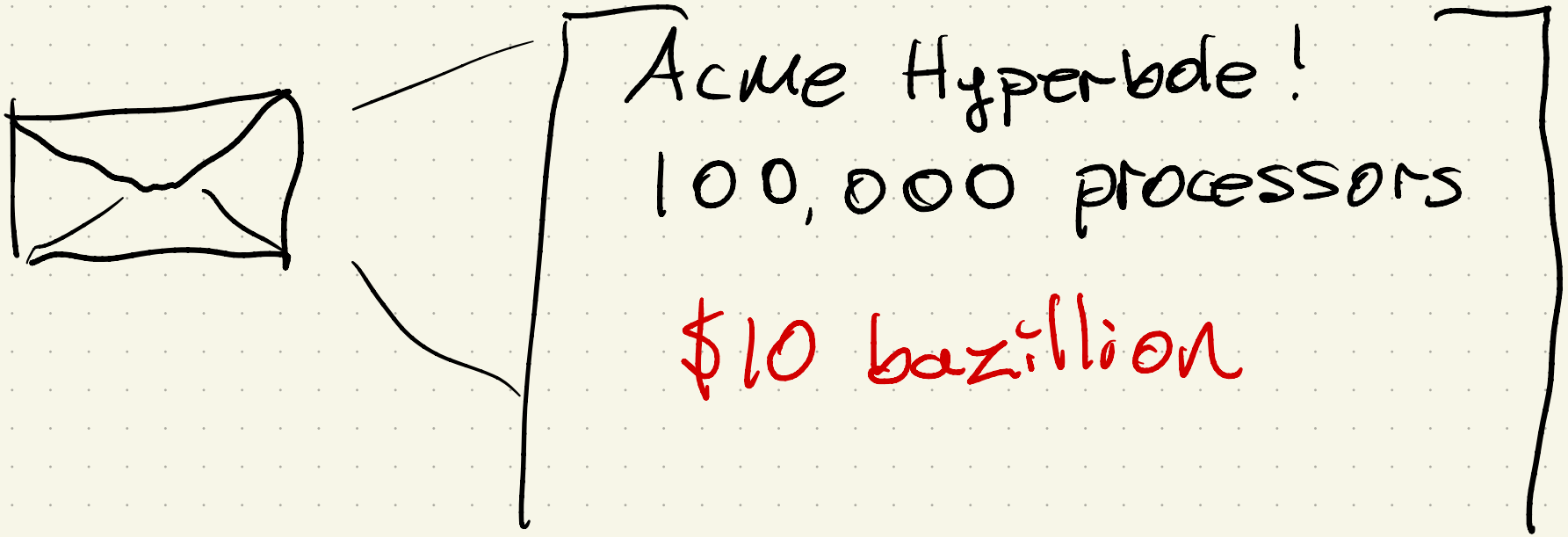


$$\text{speedup} = \frac{\text{Time}_1}{\text{Time}_N} = \frac{\text{Time}_1}{(1-F) \times \text{Time}_1 + F \times \left(\frac{\text{Time}_1}{N}\right)}$$

$$\text{Speedup} = \frac{1}{(1-F) + \frac{F}{N}}$$ Amdahl's law

Boeing

1 run takes 1 week
on fastest computer



1 week ≈ 600,000 sec, 1 sim in 6 sec!

Finding f: 80% of code is parallelizable!

$$\text{Speedup} = \frac{1}{(1-f) + f/N}$$

$$= \frac{1}{(1-0.8) + \frac{0.8}{100000}}$$

$$= \frac{1}{(1-0.8)} = \frac{1}{0.2} = 5$$

Speedup is limited by sequential work

1 sim will take ~ 33 hours

Say f of 90%: $\frac{1}{(1-0.9)} = 10$

99%: $\frac{1}{1-0.99} = 100$

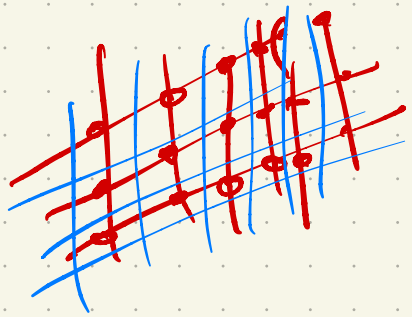
Gustafson: "soft scaling"

$$\text{Speedup} = \frac{T1 + T2 \times N}{T1 + T2}$$

α = sequential fraction of parallel program

$$= \frac{T1}{T1 + T2}$$

← sequential time in a parallel program



$$\text{Speedup} = \alpha + (1-\alpha) * N$$

$$= N - \alpha(N-1)$$

Amdahl's $f \neq$ Gustafon's α

$f \neq 1 - \alpha$

Combining #s

	Computer A	B	C
Prog P1	1 sec	10 sec	20 sec
Prog P2	1000 sec	100 sec	20 sec

Only P1 matters? Buy computer A

only P2 matters? Buy computer C

Both matter equally?

run P1 then P2

	A	B	C
	1001	110	40
Average	500.5	55	20

	X	Y
Prog P1	4	2
Prog P2	4	7
Ave	4	4.5

← Transactions per second

If both P1 and P2, which one?

Average suggests by Y

Time for 1 transaction

	X	Y
Prog P1	1/4	1/2
Prog P2	1/4	1/7
Total	1/2	9/14
Ave	1/4	9/28 sec
rate	4	28/9 (3.11 trans/sec)

Conclusion: buy X!

Averaging rates

$$\text{Harmonic mean} = \frac{1}{\frac{1}{N}\sum_i \frac{1}{\text{rate}_i}} = \frac{N}{\sum_i \frac{1}{\text{rate}_i}}$$

Use when bigger # is better

vs.

$$\text{Arithmetic mean} = \frac{1}{N}\sum_i \text{time}_i$$

Use when lower # is better

Geometric mean (don't use this)

$$= \left(\prod_i x_i \right)^{1/n}$$